

Exercise 1a - Hydrological Response Units

- > Divide the catchment area into smaller units representing a similar hydrological behavior.
 - Work in groups
 - Define your different types of HRUs and provide some basic properties.
 - HRUs do not need to be continuous in space
- > Data
 - swissTLMRegio: <https://www.swisstopo.admin.ch/en/landscape-model-swisstlmregio>
 - Hydrography/swissTLMRegio_FlowingWater.shp
 - Landcover/swissTLMRegio_LandCover.shp
 - DEM from Swisstopo

Exercise 1b - Perceptual model

- > You aim at creating a conceptual model for modelling the water balance of the catchment. The water balance equation is:

$$\frac{dS}{dt} = P - E - G - Q \text{ [mm/d]}$$

dS/dt = change in water storage

P = precipitation [mm/d]

E = evapotranspiration [mm/d]

G = loss to groundwater (deep percolation) [mm/d]

Q = streamflow [mm/d]

Create a perceptual model in the form of a labelled flowchart containing the processes you can think of, with some explanations.

Ex: <https://hess.copernicus.org/articles/9/95/2005/hess-9-95-2005.html>

Submission and deadline

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- > Submit 1 pdf per group (with all your names on it) on Ilias
 - > Deadline: to be communicated